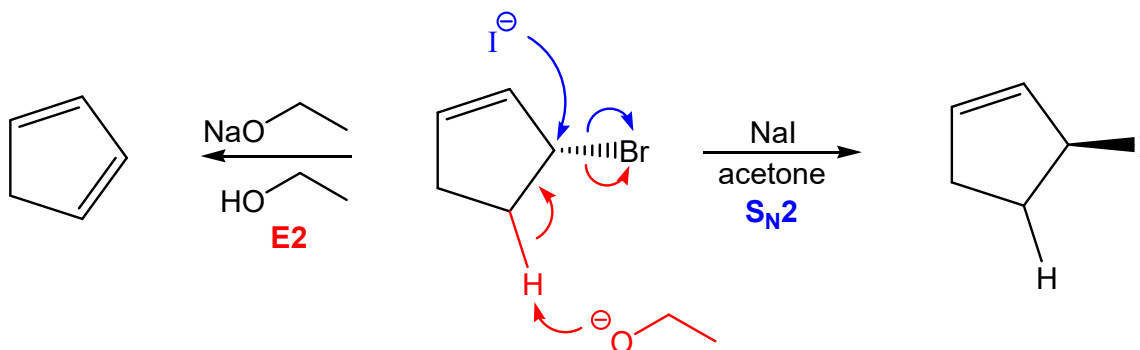
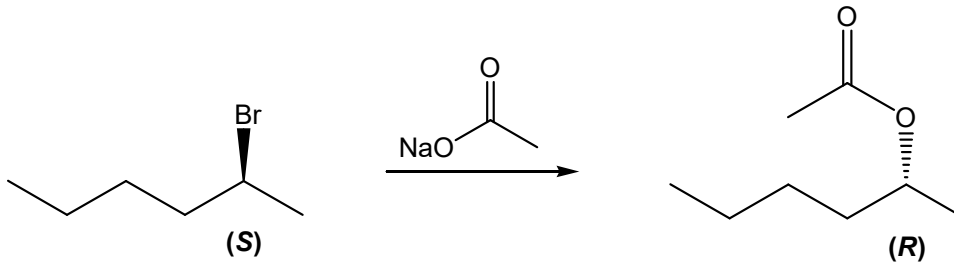


Nucleophilic Substitution and β -Elimination Worksheet Key

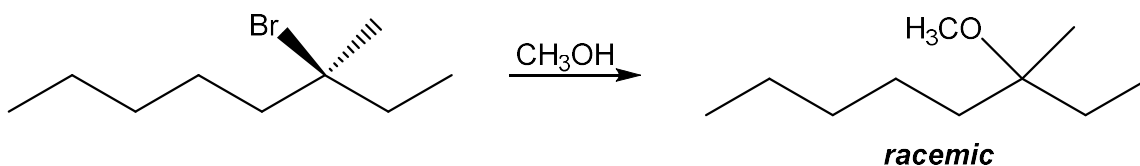
1. Draw a mechanism for the following reactions:



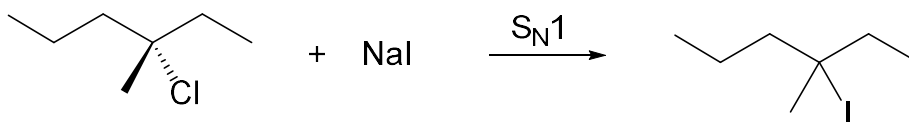
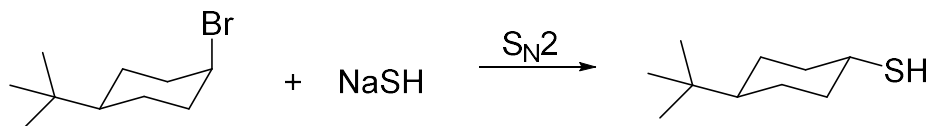
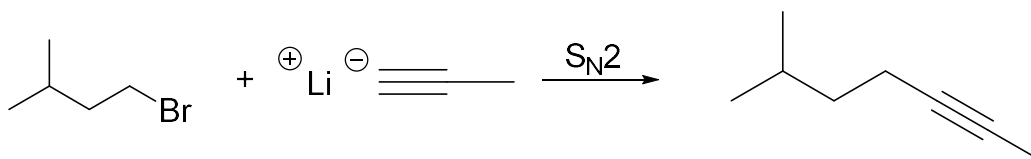
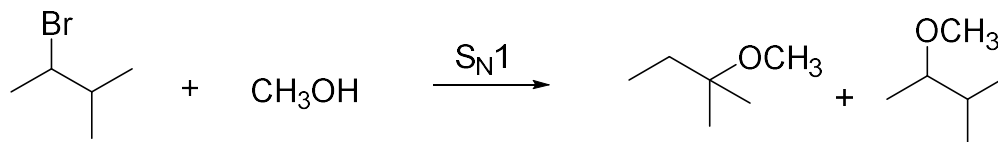
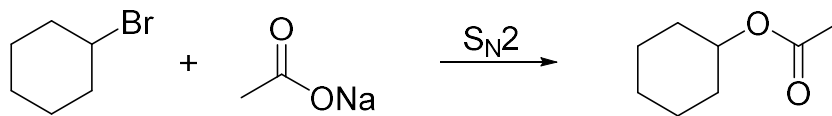
2. What product results from reaction of (S)-2-bromohexane and sodium acetate (NaOOCCH_3)? Draw the **reactant** and the **product** and indicate stereochemistry of **both**.



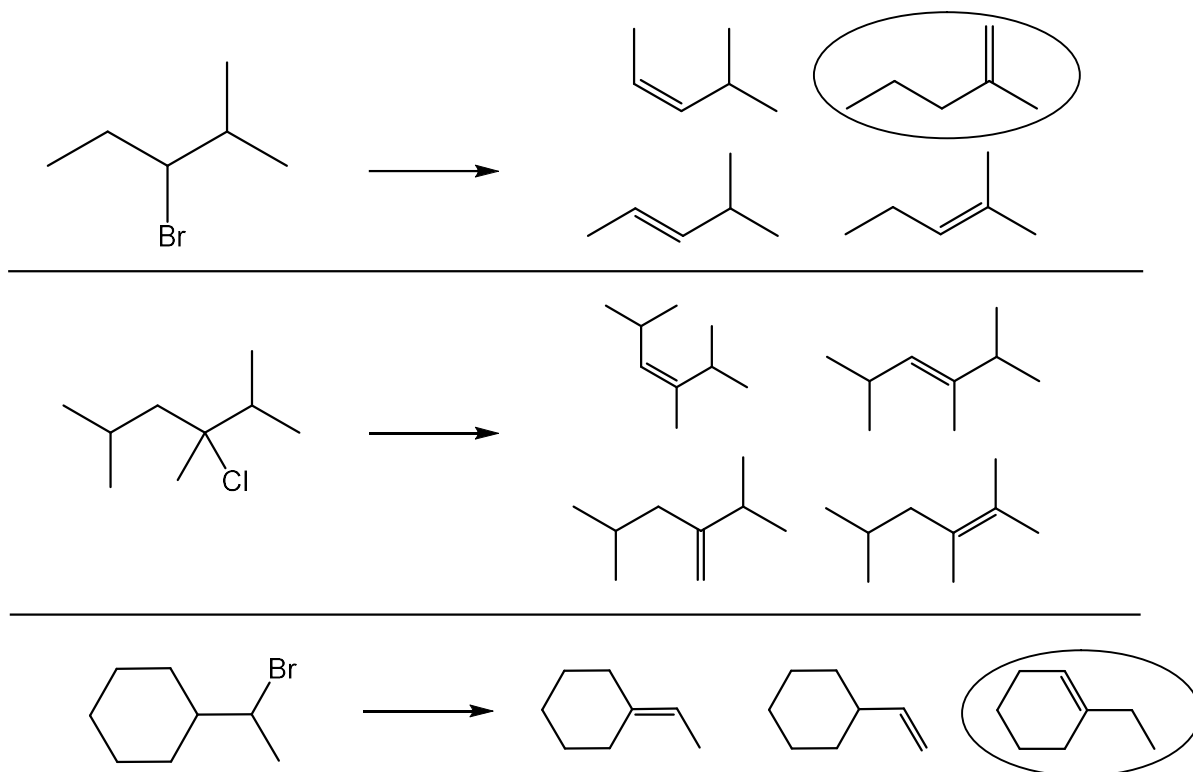
3. What *substitution* product would result from the solvolysis of (S)-3-bromo-3-methyloctane in methanol? Draw the **reactant** and the **product** and indicate stereochemistry of **both**.



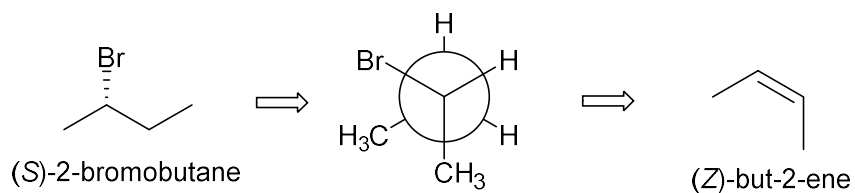
4. Draw the major organic product of the following reactions and identify whether the reaction proceeded through an S_N1 or an S_N2 mechanism based on the degree of substitution of the substrate and the strength of the nucleophile.



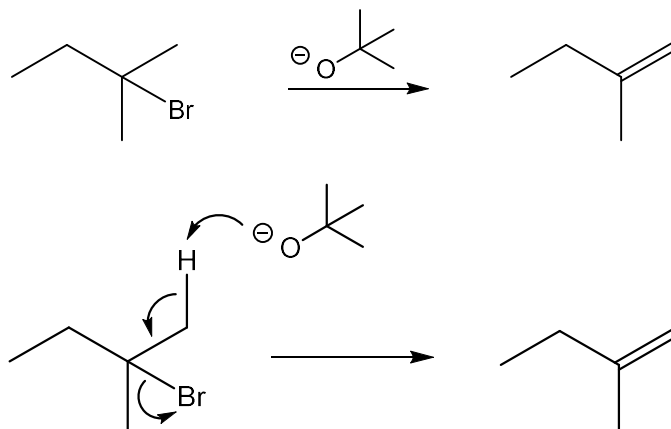
5. Draw **ALL** elimination products expected from the following alkyl halides. Circle those which can only be generated from an E1 mechanism.



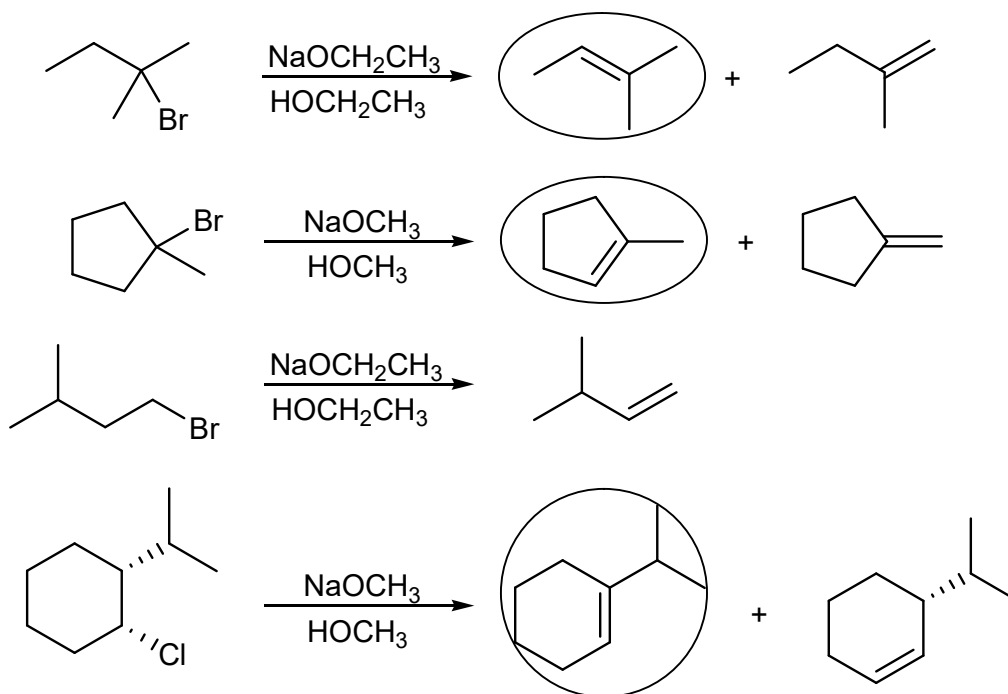
6. Sighting down C2 and C3 of (S)-2-bromobutane, draw a Newman projection that shows the proper orientation required for elimination to (Z)-but-2-ene.



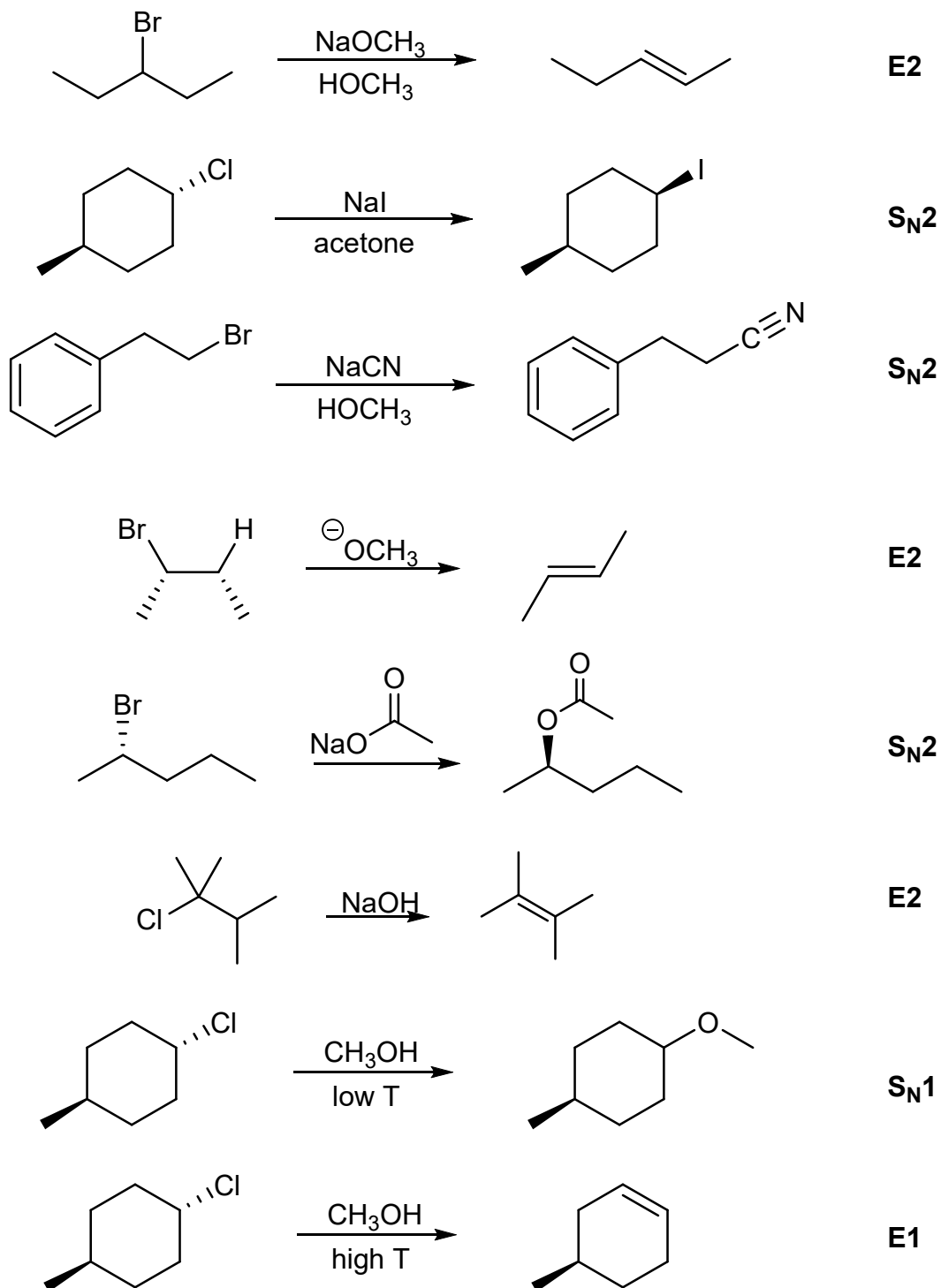
7. Draw a mechanism for the following reaction.



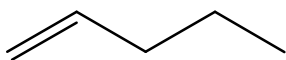
8. Draw the elimination products for the following and if a mixture of alkene products is predicted, circle the preferred product.



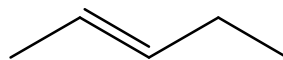
9. Draw the major organic product and predict whether each reaction proceeds by S_N1, E1, S_N2, or E2.



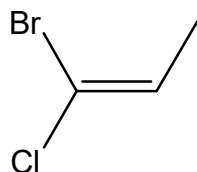
10. Write the IUPAC name for the following molecules.



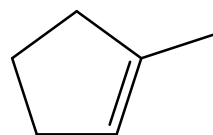
pent-1-ene



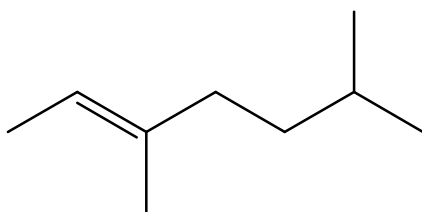
(*E*)-pent-2-ene



(*Z*)-1-bromo-1-chloroprop-1-ene

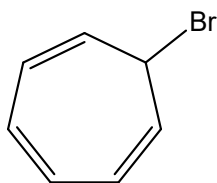


1-methylcyclopent-1-ene

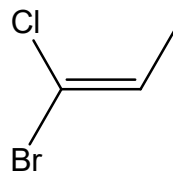


(*E*)-3,6-dimethylhept-2-ene

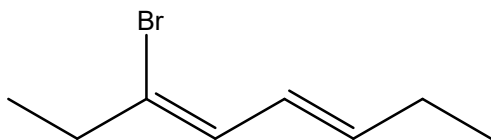
11. Draw the line structure that corresponds to the following IUPAC names.



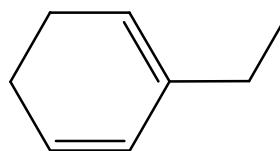
7-bromocyclohepta-1,3,5-triene



(*E*)-1-bromo-1-chloroprop-1-ene



3-bromo-(3*Z*,5*E*)-octa-3,5-diene



2-ethylcyclohexa-1,3-diene